B Evaluation metrics

For a K-class classification problem, with N samples, NLL and Brier score are calculated as $-N^{-1}\sum_{n=1}^{N}\sum_{k=1}^{K} y_{i,k}\cdot\ln\left(p_{i,k}\right)$ and $-K^{-1}N^{-1}\sum_{n=1}^{N}\sum_{k=1}^{K}(y_{i,k}-p_{i,k})^2$, respectively. Where $y_{i,k}$ is the true one-hot encoded label which is 1 if sample i has label $k\in K$, and otherwise is 0. $p_{i,k}$ is the predicted class probability of sample i belonging to class $k\in K$. Reliability diagrams plot expected accuracy as a function of class probability (confidence). Expected Calibration Error (ECE) is used to summarize the results of reliability diagrams. Details of evaluation metrics are given in the Supplementary Material. For expected accuracy measurement, the samples are binned into N groups and the accuracy and confidence for each group are computed. Assuming D_m to be indices of samples whose confidence predictions are in the range of $\left(\frac{m-1}{M},\frac{m}{M}\right]$, the expected accuracy of the D_m is $Acc(D_m)=|D_m|^{-1}\sum_{i\in D_m} y_{i,k}$. The average confidence on bin D_m is calculated as $\overline{P}(D_m)=|D_m|^{-1}\sum_{i\in D_m} p_{i,k}$. ECE is calculated by summing up the weighted average of the differences between accuracy and the average confidence over the bins: ECE $=\sum_{m=1}^{M} N^{-1}|D_m|\left|ACC(D_m)-\overline{P}(D_m)\right|$.

C Additional Results on ImageNet

Figure 1 shows reliability diagrams together with ECE values for baseline, temperature scaling and parameter ensembling with perturbation (PEP) for the pre-trained ImageNet networks.

References

[1] Arakaparampil M Mathai and Serge B Provost. *Quadratic forms in random variables: theory and applications*. Dekker, 1992.

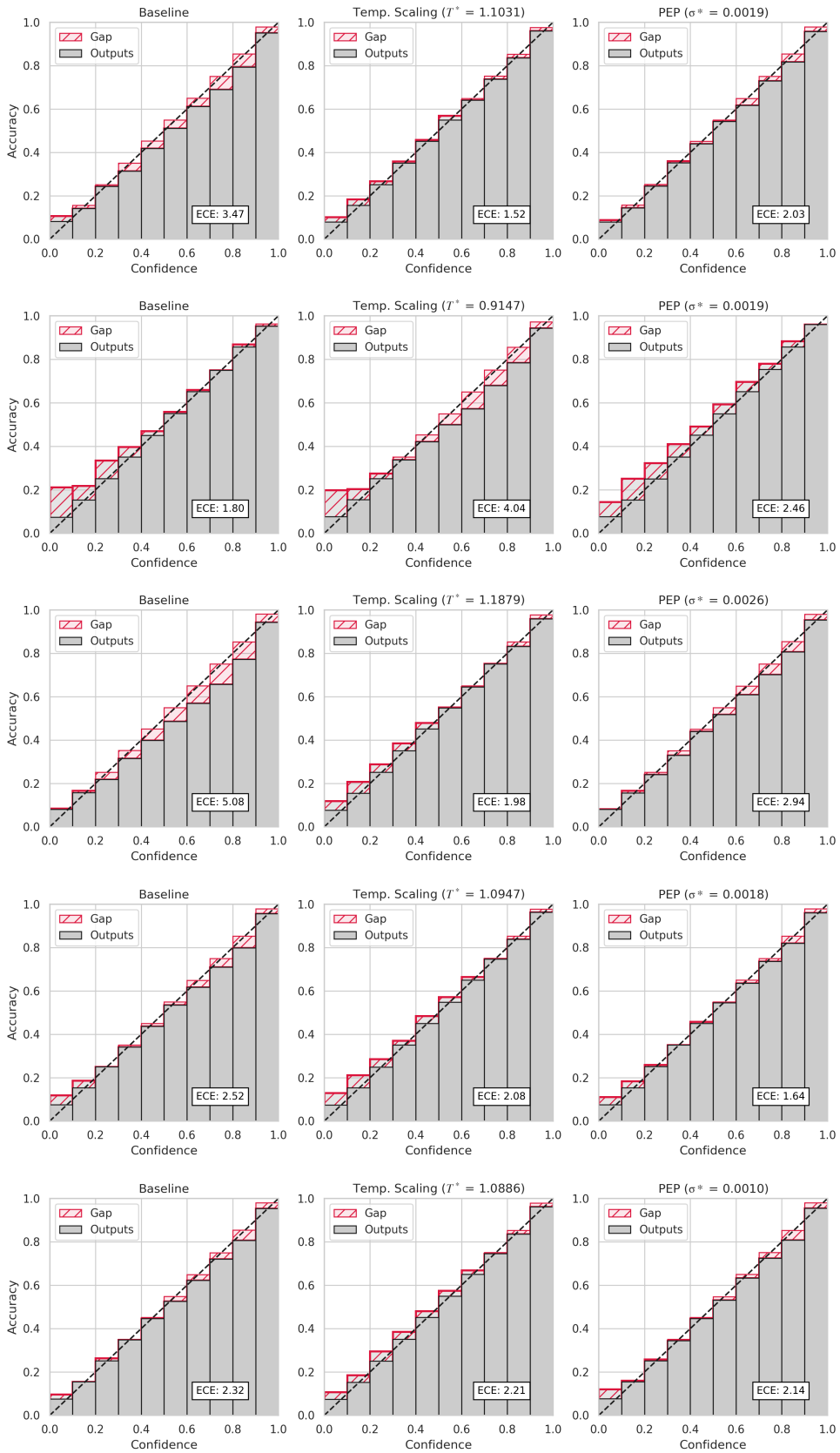

Figure 1: Reliability diagrams and ECE values before and after calibration with Temperature Scaling and PEP, for experiments described in Section 3.1 of the manuscript. From top to bottom: DenseNet121, InceptionV3, ResNet50, VGG16, and VGG19.